KUBERNETES HOMELAB

Technical Project Overview

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Platform Engineering | DevOps | Cloud Infrastructure

GitHub: github.com/mmrajput/kubernetes-homelab

1. Project Objective

This homelab project demonstrates production-ready platform engineering through hands-on implementation on a 3-node Kubernetes cluster running on Proxmox VE, mirroring enterprise deployment patterns from infrastructure provisioning through application delivery.

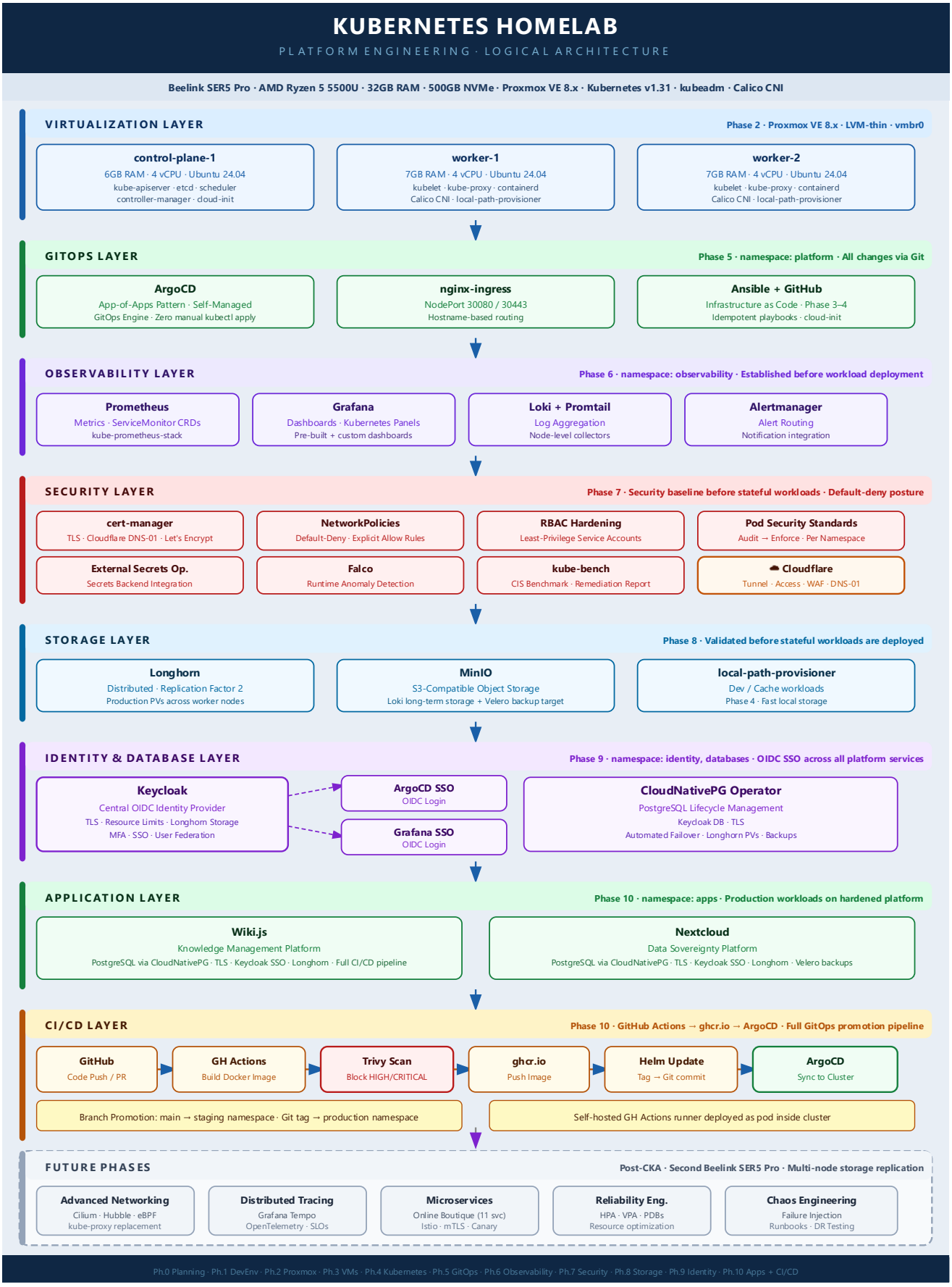
The project follows a deliberate sequence that reflects real platform engineering priorities: build the platform, secure it, establish identity management, then deploy production-grade workloads on a hardened foundation. GitOps-first operations via ArgoCD, infrastructure automation via Ansible, and full observability through Prometheus, Loki, and Grafana are established before any application workloads are introduced.

Security is treated as a foundation rather than an afterthought — TLS via cert-manager, default-deny NetworkPolicies, Pod Security Standards, RBAC hardening, and Falco runtime security are all in place before stateful workloads are deployed. Keycloak on CloudNativePG-managed PostgreSQL serves as the central identity provider with OIDC integration across all platform services.

The platform is validated through two real-world application deployments. Wiki.js confirms the end-to-end CI/CD pipeline and demonstrates the complete platform stack. Nextcloud — a core component of the German government's Sovereign Workplace initiative — demonstrates data sovereignty principles, running with Keycloak SSO, CloudNativePG-managed PostgreSQL, TLS throughout, and automated backups via Velero. Both applications are deployed through an automated GitHub Actions to ArgoCD pipeline. All architectural decisions are documented as Architecture Decision Records.

The full platform architecture diagram is available on the next page and in the project repository at [github.com/mmrajput/kubernetes-homelab/tree/main/resources/images/platform-architecture.svg]

2. Platform Architecture



3. Phase Details

Phase 0: Architecture Design & Planning (Complete)

Established the architectural foundation through systematic planning and documentation before writing a single line of configuration.

- Architecture Decision Records (ADRs) covering hardware selection, hypervisor choice, Kubernetes distribution, CNI, and GitOps strategy with trade-off analysis for each decision
- Capacity planning for 32GB RAM across 3 VMs with headroom for workload expansion
- Network topology design with static IP allocation strategy

Future Enhancements:

- README updates and documentation maintained continuously across all phases

Phase 1: Development Environment (Complete)

Reproducible development environment using DevContainers with version-pinned tooling, ensuring consistent operations across sessions.

- Ubuntu 24.04 base image matching cluster nodes
- Pinned versions: kubectl v1.31.4, helm v3.16.3, k9s v0.32.7, argocd CLI
- Mounted kubeconfig for seamless cluster access

Phase 2: Virtualization Foundation (Complete)

Deployed Proxmox VE 8.x as the Type-1 hypervisor on a Beelink SER5 Pro mini-PC. Configured storage, network bridging, and virtualization foundation for all cluster nodes.

- Hardware: AMD Ryzen 5 5500U, 32GB RAM, 500GB NVMe
- Linux bridge (vmbr0) for VM network connectivity
- LVM-thin provisioning for efficient storage allocation

Phase 3: VM Provisioning (Complete)

Automated virtual machine provisioning using cloud-init templates for repeatable, consistent and scalable cluster node deployment.

- Ubuntu 24.04 cloud image template with cloud-init integration
- Automated provisioning workflows for control plane and worker nodes
- Deterministic static IP assignment and SSH key injection

Future enhancements

- Terraform-based IaC for VM lifecycle management

Phase 4: Kubernetes Cluster (Complete)

Deployed production-grade Kubernetes v1.31 using kubeadm with Ansible automation. Calico CNI provides pod networking with NetworkPolicy support.

- 3-node cluster: 1 control plane (6GB), 2 workers (7GB each)
- Idempotent Ansible playbook for repeatable cluster deployments

- Calico CNI selected over Flannel for NetworkPolicy enforcement capability
- local-path-provisioner for dynamic PersistentVolume (PV) provisioning

Phase 5: GitOps & Platform Services (Complete)

GitOps workflow using ArgoCD with App-of-Apps pattern. All platform services managed declaratively through Git, no manual kubectl apply in production.

- ArgoCD with App-of-Apps pattern for scalable multi-application management
- ArgoCD self-manages its own configuration via Git commits
- nginx-ingress with NodePort (30080/30443) for service exposure
- Hostname-based routing established for all platform services

Future enhancements

- Forgejo self-hosted Git server deployed as a Kubernetes workload, enabling a fully self-contained GitOps workflow independent of external services

Phase 6: Observability Stack (Complete)

Full metrics, logging, and visualization pipeline following cloud-native best practices. Observability is established before complex workloads are deployed.

- Prometheus for metrics collection with ServiceMonitor CRDs
- Grafana for visualization with pre-built Kubernetes dashboards
- Loki for log aggregation with Promtail node collectors

Phase 7: Security Hardening (Complete)

Security baseline established before any stateful workloads are deployed. All services currently running over HTTP, closing this gap is the primary objective of this phase.

- cert-manager with Cloudflare DNS-01 ClusterIssuer for Let's Encrypt wildcard certificate issuance, no inbound internet exposure required
- TLS termination at nginx-ingress for all platform services (ArgoCD, Grafana, Prometheus)
- NetworkPolicies across all namespaces, default-deny with explicit allow rules
- Pod Security Standards enforced per namespace, audit mode first, then enforce
- RBAC hardening, least-privilege service accounts, removal of unnecessary cluster-admin bindings

Future enhancements

- External Secrets Operator for secrets backend integration – After Phase 8
- kube-bench CIS benchmark run and remediation report – After Phase 10
- Falco runtime security for anomaly detection in running containers – After Phase 10

Phase 8: Storage & Backup (In Progress)

Distributed storage and disaster recovery infrastructure established before stateful workloads are deployed. Storage must be validated before it is trusted with production data.

- Longhorn distributed storage across worker nodes with replication factor 2
- MinIO S3-compatible object storage, single deployment serving both Loki long-term storage and Velero backup target

- Storage hierarchy: local-path (dev/cache), Longhorn (stateful production), MinIO (object/backup)

Future enhancements

- Velero for Kubernetes-native backup and restore with scheduled backup policies
- Documented DR procedures with tested restore playbook

Phase 9: Stateful Applications & Identity (Planned)

Deploy PostgreSQL and Keycloak as the stateful foundation that production applications in Phase 10 depend on. Keycloak become the central identity provider for all platform services.

- CloudNativePG operator for PostgreSQL lifecycle management
- Keycloak deployed on CloudNativePG with TLS, resource limits, and Longhorn storage
- Keycloak as OIDC provider for ArgoCD and Grafana SSO
- Homepage dashboard for internal service discovery.

Phase 10: CI/CD Pipeline & Applications (Planned)

Complete GitHub Actions to ArgoCD pipeline demonstrated through real-world application deployments.

- GitHub Actions workflow triggered on push to main branch
- Docker image build using GitHub Actions native build infrastructure
- Trivy vulnerability scan — pipeline blocks on HIGH/CRITICAL CVEs
- Image pushed to GitHub Container Registry (ghcr.io)
- Automated image tag update committed back to Helm values repository
- ArgoCD detects Helm chart change and syncs deployment to cluster
- GitHub Actions self-hosted runner deployed in cluster for Kubernetes-native build steps
- Branch-based promotion — main branch deploys to staging namespace, Git tags promote to production namespace
- Pull Request preview environments — ephemeral namespace per PR, destroyed on merge
- Separate repositories for application source code and Helm charts following GitOps convention

Application Deployments:

- **Wiki.js** is deployed first as the pipeline validation workload — a straightforward PostgreSQL-backed application that confirms the end-to-end pipeline work and demonstrates the complete platform stack: TLS, Keycloak SSO, CloudNativePG database management, Longhorn persistent storage, and automated GitOps deployment.

- **Nextcloud** is deployed second as the primary production workload, demonstrating data sovereignty principles directly relevant to German public sector IT and the Sovereign Workplace initiative. It introduces additional platform complexity — Redis caching, RWX persistent storage, PgBouncer connection pooling, and CronJob background processing — making it a more demanding operational challenge than Wiki.js.

Future Phase (Advanced Post-CKA)

These phases are planned following CKA certification and hardware expansion. A second or third Mini-PC will be added as an additional node(s), enabling true multi-node storage replication, workload distribution across physical hosts, and realistic failure scenario testing that a single-host setup cannot provide.

- **Advanced Networking**: Calico to Cilium migration with Hubble observability and eBPF-based networking
- **Distributed Tracing:** Grafana Tempo, OpenTelemetry Collector, and SLO/error budget dashboards
- **Microservices & Service Mesh:** Google Online Boutique (11-service, polyglot architecture) as the reference microservices workload, with Istio for mTLS east-west encryption, traffic management patterns including canary deployments and circuit breaking, and distributed tracing across all services
- **Reliability Engineering:** HPA, VPA, PodDisruptionBudgets, and resource optimization
- **Chaos Engineering:** Structured failure injection with documented runbooks and recovery playbooks

4. Technology Stack

Implementation Status:

- **Implemented:** Phases 0–6 complete · 15 technologies deployed and operational
- **In Progress:** Phase 7 active · Security hardening and TLS rollout underway
- **Planned:** Phases 8 – 10 · Storage, Identity, CI/CD, and Application workloads
- **Advanced (Post-CKA):** Cilium, Istio, distributed tracing, chaos engineering · planned after CKA certification.

Layer	Technology	Purpose	Phase	Status
Infrastructure	**Proxmox VE 8.x**	Type-1 hypervisor · LVM-thin · vmbr0 bridge	2	**Implemented**
Infrastructure	**Ubuntu 24.04 LTS**	VM operating system · cloud-init provisioned nodes	3	**Implemented**
Infrastructure	**Cloud-init**	Automated VM provisioning · SSH key injection	3	**Implemented**
Infrastructure	**Ansible**	Infrastructure as Code · idempotent cluster deployment	4	**Implemented**
Kubernetes Core	**kubeadm**	Cluster bootstrap · control plane management	4	**Implemented**
Kubernetes Core	**containerd**	Container runtime (CRI)	4	**Implemented**
Kubernetes Core	**Calico CNI**	Pod networking · NetworkPolicy enforcement	4	**Implemented**
Kubernetes Core	**local-path-provisioner**	Dynamic PV provisioning · dev/cache workloads	4	**Implemented**
GitOps	**ArgoCD**	GitOps engine · App-of-Apps pattern · self-managed	5	**Implemented**
GitOps	**nginx-ingress**	HTTP/HTTPS routing · NodePort 30080/30443	5	**Implemented**
Observability	**Prometheus**	Metrics collection · ServiceMonitor CRDs · alerting	6	**Implemented**
Observability	**Grafana**	Dashboards · Kubernetes panels · visualization	6	**Implemented**
Observability	**Loki + Promtail**	Log aggregation · node-level collectors	6	**Implemented**

Layer	Technology	Purpose	Phase	Status
Security	**cert-manager**	TLS automation · Cloudflare DNS-01 · Let's Encrypt wildcard	7	**Implemented**
Security	**NetworkPolicies**	Default-deny posture · explicit allow rules per namespace	7	**Implemented**
Security	**RBAC Hardening**	Least-privilege service accounts · cluster-admin removal	7	**Implemented**
Security	**Pod Security Standards**	Audit then enforce per namespace	7	**Implemented**
Security	**External Secrets Op.**	Secrets backend integration · no plaintext secrets in Git	7	**In Progress**
Security	**Falco**	Runtime anomaly detection · syscall-level monitoring	7	**In Progress**
External Access	**Cloudflare**	Tunnel · Access (zero-trust) · WAF · DNS-01 challenge	7	**Implemented**
Storage	**Longhorn**	Distributed PVs across workers · replication factor 2	8	**In Progress**
Storage	**MinIO**	S3-compatible · Loki long-term storage + Velero backup target	8	**In Progress**
Storage	**Velero**	Kubernetes-native backup and restore · scheduled policies	8	**In Progress**
Identity & DB	**Keycloak**	Central OIDC provider · SSO for ArgoCD and Grafana	9	**Planned**
Identity & DB	**CloudNativePG**	PostgreSQL operator · automated failover · TLS · backups	9	**Planned**
Applications	**Wiki.js**	Knowledge management · PostgreSQL · TLS · CI/CD validation workload	10	**Planned**
Applications	**Nextcloud**	Data sovereignty · file management · Sovereign Workplace initiative · Keycloak SSO	10	**Planned**
CI/CD	**GitHub Actions**	Pipeline trigger · Docker image build · self-hosted runner in cluster	10	**Planned**
CI/CD	**Trivy**	Container image vulnerability scanning · block on HIGH/CRITICAL	10	**Planned**
CI/CD	**ghcr.io**	GitHub Container Registry · image storage and versioning	10	**Planned**
Post-CKA	**Cilium**	Calico migration · eBPF networking · Hubble observability	11+	**Advanced**
Post-CKA	**Grafana Tempo**	Distributed tracing · OpenTelemetry · SLO dashboards	11+	**Advanced**
Post-CKA	**Istio**	Service mesh · mTLS east-west encryption · traffic management	12+	**Advanced**
Post-CKA	**Chaos Mesh**	Structured failure injection · runbook validation	14+	**Advanced**
Post-CKA	**Microservices & Service Mesh**	Online Boutique (11-service polyglot) as reference workload · gRPC · Istio mTLS · distributed tracing	12+	**Advanced**

5. Certification & Career Roadmap

This homelab is deliberately structured to build toward CNCF certification and platform engineering roles. Every phase produces hands-on experience that maps directly to certification exam topics and real job requirements.

Immediate Goal: CKA Certification

The Certified Kubernetes Administrator (CKA) exam tests practical cluster administration under time pressure — precisely the skills this homelab builds through kubeadm cluster deployment, Calico CNI configuration, persistent volume management, RBAC, and real troubleshooting scenarios. Phases 0–7 cover the majority of the CKA curriculum through actual implementation rather than exam preparation alone.

Next: CKS Certification

The Certified Kubernetes Security Specialist (CKS) extends CKA with a security-first mindset. Phase 7 of this homelab — NetworkPolicies with default-deny posture, Pod Security Standards, RBAC hardening, Falco runtime security, and kube-bench CIS benchmarking — is structured to align directly with the CKS curriculum, making the certification a natural next step after CKA rather than a separate preparation effort.

Career Target

Targeting DevOps and Platform Engineering roles in the Stuttgart/Baden-Württemberg region, with interest in government IT, manufacturing, mobility, energy, and e-commerce sectors. German B2 language certification obtained. Actively seeking roles from Q2 2026.

6. Repository & Contact

GitHub Repository: github.com/mmrajput/kubernetes-homelab

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